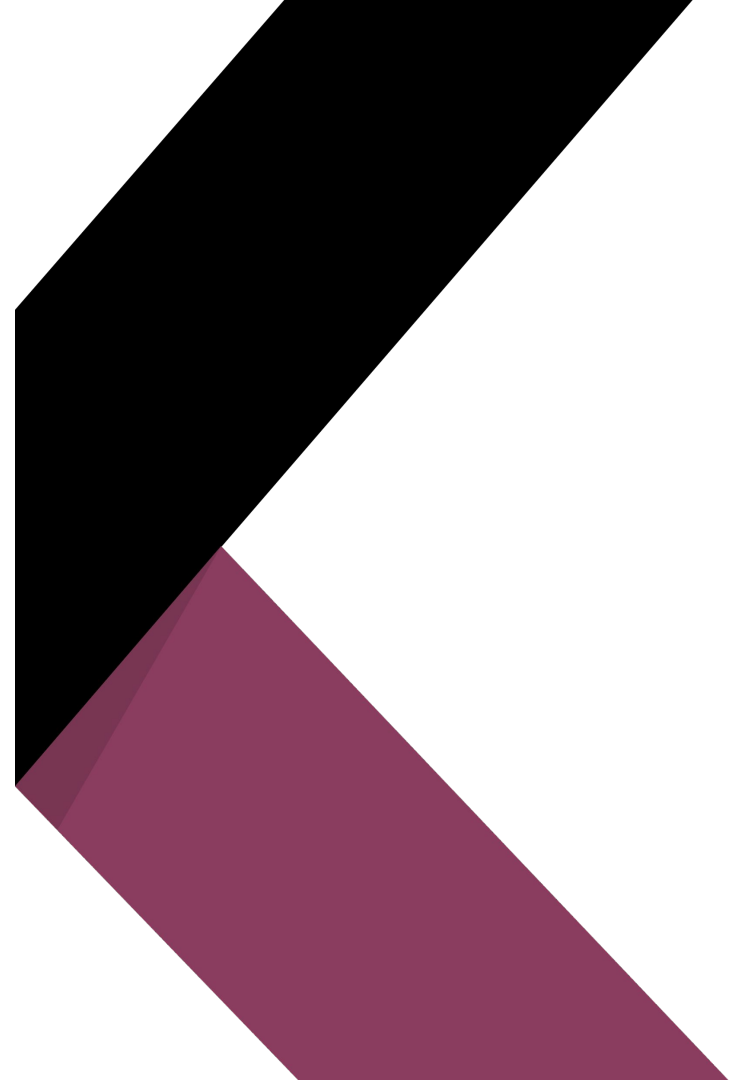


Diagnosis

CDA

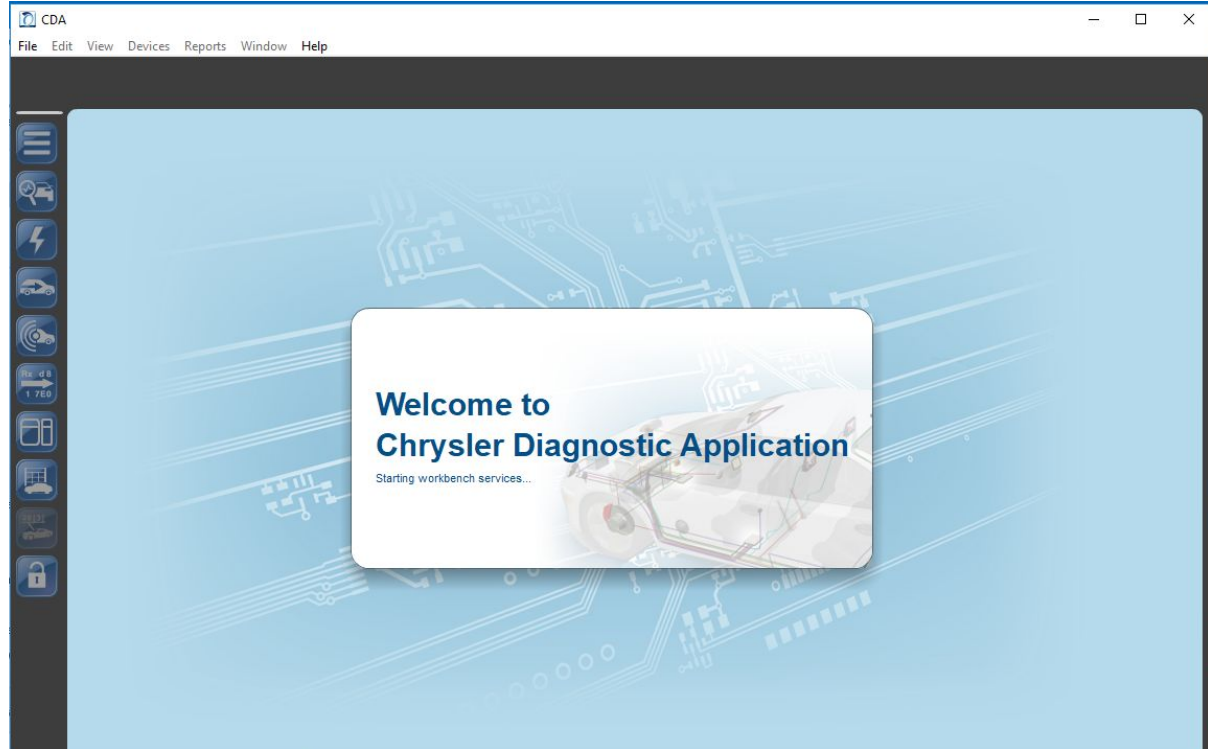




CDA – Chrysler Diagnostic Application

CDA is the diagnostic tool of Chrysler; at the moment FCA is using this tool on some projects.

CDA is quite similar and equivalent to DIAnalyzer.

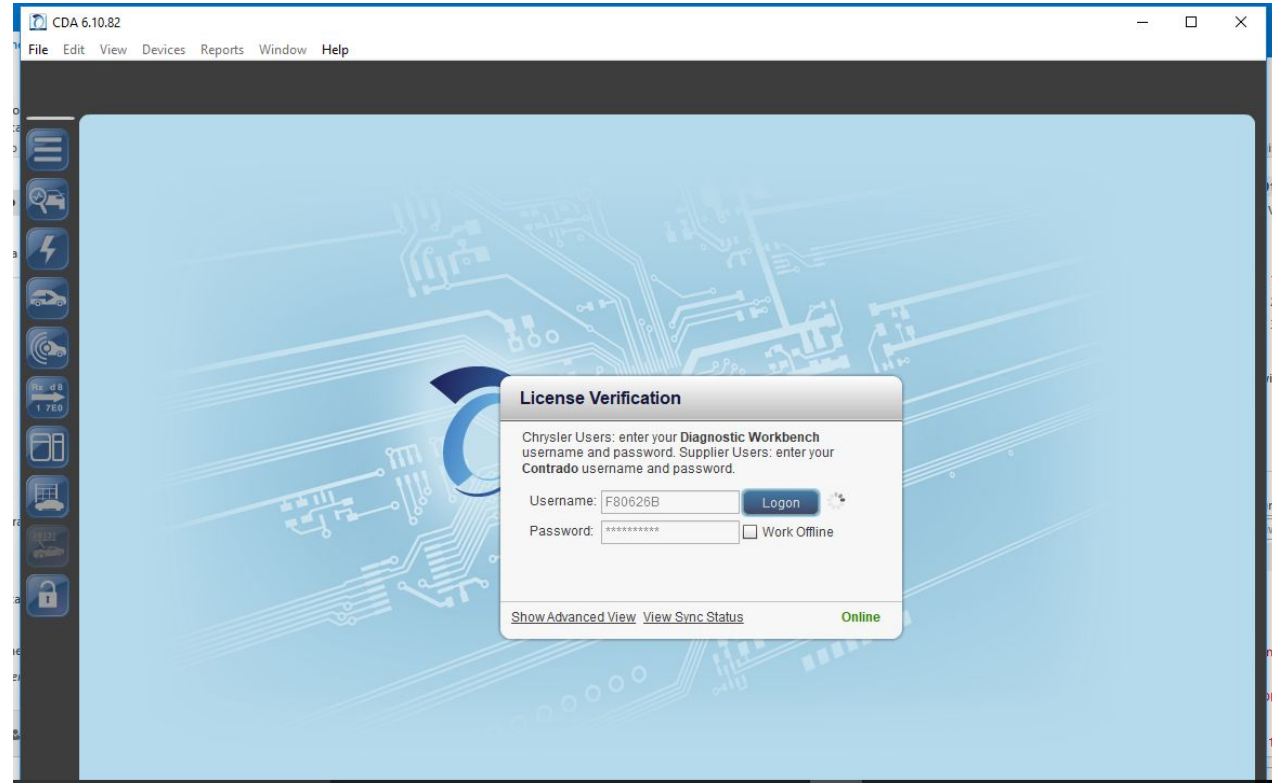




CDA – Chrysler Diagnostic Application

Any user needs of a specific enabled ID to use CDA.

It's possible to use CDA in Offline (check «Work Offline») but it's necessary a specific user privilege.

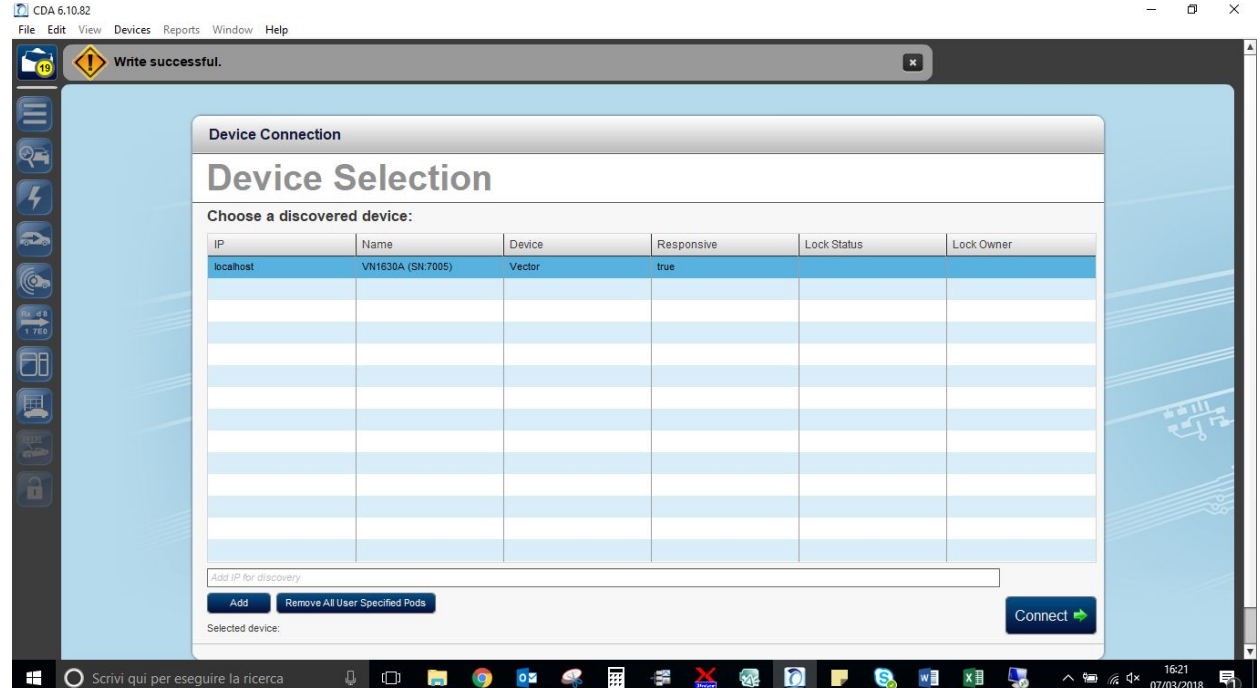




CDA – Device selection

The first step to exchange diagnostic data with CDA is the selection of the hardware device: in this screen it's necessary select the device and press the button Connect.

Pay attention that CDA can connect itself only to known hardware; the hardware needs to be installed correctly on the host PC.





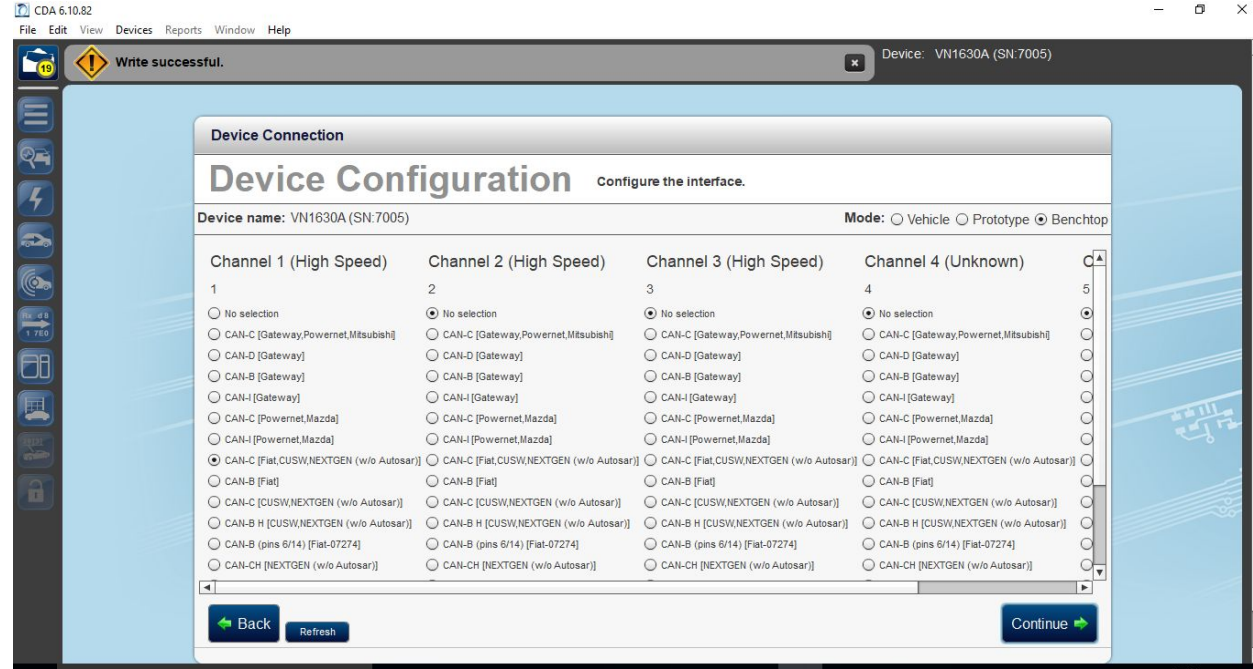
CDA – Device configuration

Device configuration

CDA needs to know if the user is working with a «Vehicle» or with a simulated environment («Benchtop»)

Channel assignment:

The user has to select the channel that he wants to use to establish the diagnostic communication between ECU and CDA





CDA – ECU selection

ECU selection.

The user has to select the ECU for the diagnostic communication. It's necessary scroll, select and double click on the selected ECU

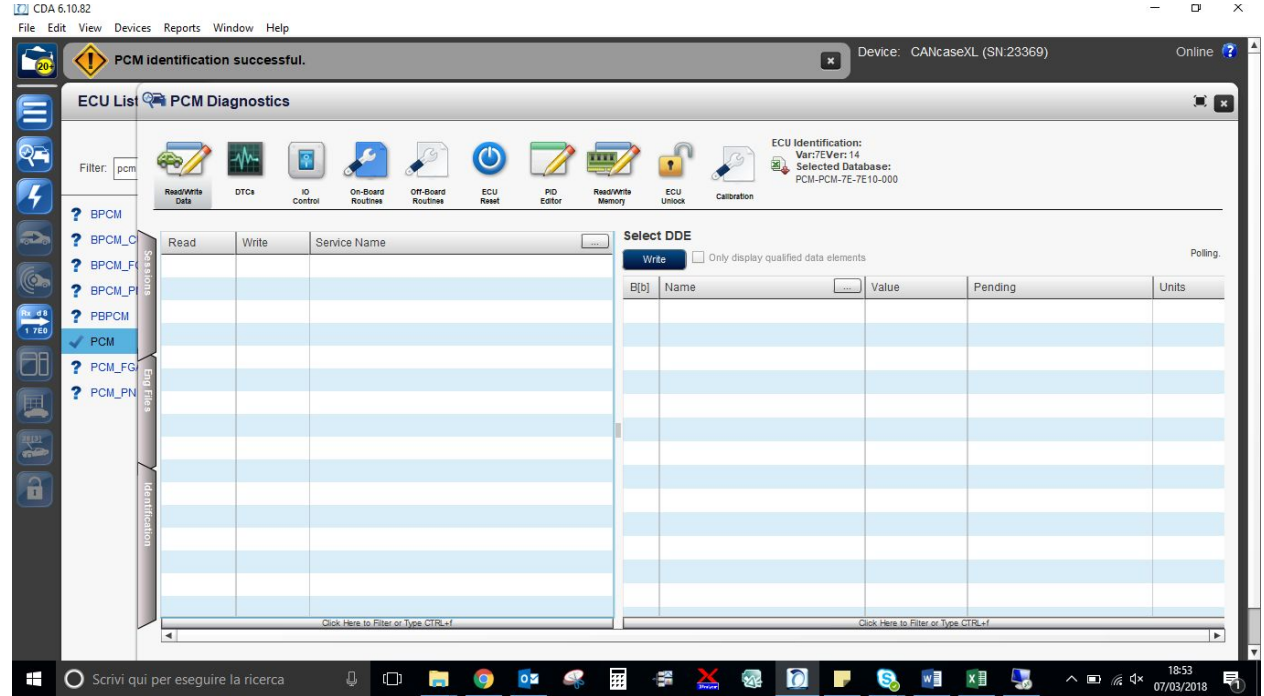




CDA – Verification of the ECU connection

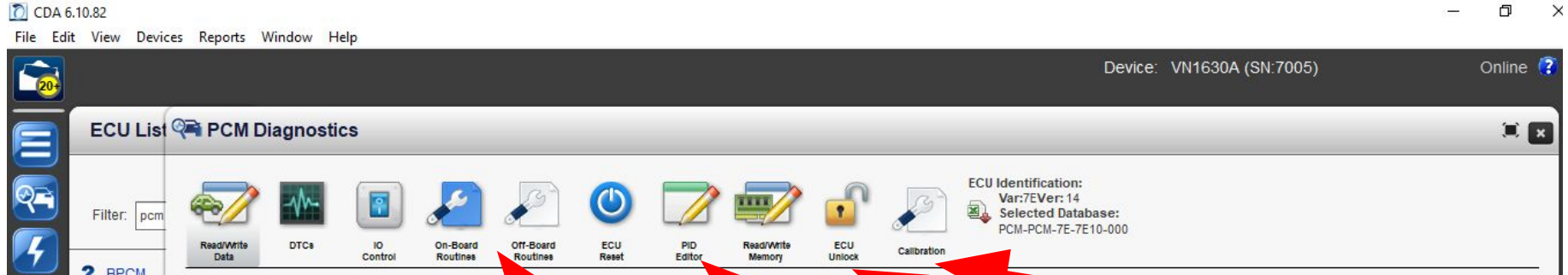
In case of a successful connection with the selected ECU:

- It appears a blue tick near the selected ECU;
- It appears the message related to the «Identification successful».
- CDA opens the diagnostic windows of the selected ECU
- Now it's possible to navigate among all the diagnostic windows to manage all the diagnostic functionalities





CDA – Diagnostic screen



Read/Write **RDI**

DTC section:
check of all DTC
information

IOControl:
Selection and
execution of
active diagnosis
on the actuator.

ROUTINE section:
Selection and
execution of routines
(predefined system
operation on the
ECU)

ECU UNLOCK section:
it's necessary unlock ad
ECU only with a specific
user privilege

PID EDITOR section:
It's possible to send raw
value and read raw
response for the
diagnostic information.

CALIBRATION
section:

It's possible to
send a sequence
of command to
the ECU with a
.txt file



CDA – Read RDI

Read RDI :

It's possible to read an RDI with an access to section Read/Write RDI.

With a single click on the RDI, it's possible to read the content.

CDA 6.10.82

File Edit View Devices Reports Window Help

Device: CANcaseXL (SN:23369) Online

ECU List **PCM Diagnostics**

Filter: pcm

Read/Write Data DTCs IO Control On-Board Routines Off-Board Routines ECU Reset PID Editor Read/Write Memory ECU Unlock Calibration

ECU Identification:
Var:7EVer:14
Selected Database:
PCM-PCM-7E-7E10-000

Read **Write** Service Name

Read	Write	Service Name
229E0A	2E9E0A	In-Plant Test Register 1
229E0B	2E9E0B	In-Plant Test Register 2
229E0D	2E9E0D	In-Plant Test Register 4
229E0E	2E9E0E	In-Plant Test Register 5
229E10	2E9E10	In-Plant Test Register 7
229E12	2E9E12	In-Plant Test Register 9
229E13	2E9E13	In-Plant Test Register 10
229E15	2E9E15	In-Plant Test Register 11
220F1B		EVR CONFIG
220F1C		Dual Alternator Enabled
220F1E		OBD ENGINE RUNTIME
220F9B		Number Of Power Pulses
220F9C		ENGRSTPERF ENGRSTRT TMR
221160		CAN_VE_H_CFG_C1_STATUS_REGISTER
221161		COMRX_EC_ECUCfg6_Stat
221162		COMRX_EC_Powertrain

In-Plant Test Register 1

Write ☐ Only display qualified data elements

B[b]	Name	Value	Pending	Units
3[0]	IAC Low RPM Test Status	Passed	Passed	
3[2]	Battery Charging Test Status	Passed	Passed	
3[4]	Vehicle Speed Rationality Test Status	Passed	Passed	
3[6]	Shut Down Timer "Slow" Rationality Test St	Passed	Passed	
4[0]	Powertrain Control Module Mode	In-Plant	In-Plant	
4[1]	Fuel Prime Test Complete	True	True	
4[2]	Fuel Rail Bleed Test Complete	True	True	
4[3]	Air Pump DTI Actuator Test Complete	True	True	
4[4]	DTI Actuator Test Status	Passed	Passed	
4[6]	IAC High RPM Test Status	Passed	Passed	



CDA –

Write RDI

WriteRDI

It's possible write on many RDI; this is an EEPROM writing operation. An example of writing operation on an RDI is the PROXI procedure.

To modify an RDI it's possible select the notepad, modify the value and press the button «Write».

If the operation is successful, it appears a message «Write successful».

CDA 6.10.82

File Edit View Devices Reports Window Help

Device: CANcaseXL (SN:23369) Online

ECU List

Filter: pcm

- BPCM
- BPCM_C
- BPCM_F
- BPCM_P
- PBPCM
- PCM
- PCM_FG
- PCM_PN

PCM Diagnostics

Read/Write Data DTCs IO Control On-Board Routines Off-Board Routines ECU Reset PID Editor Read/Write Memory ECU Unlock Calibration

ECU Identification: Var:7EVer: 14 Selected Database: PCM-PCM-7E-7E10-000

Read	Write	Service Name
229E0A	2E9E0A	In-Plant Test Register 1
229E0B	2E9E0B	In-Plant Test Register 2
229E0D	2E9E0D	In-Plant Test Register 4
229E0E	2E9E0E	In-Plant Test Register 5
229E10	2E9E10	In-Plant Test Register 7
229E12	2E9E12	In-Plant Test Register 9
229E13	2E9E13	In-Plant Test Register 10
229E15	2E9E15	In-Plant Test Register 11
220F1B		EVR CONFIG
220F1C		Dual Alternator Enabled
220F1E		OBD ENGINE RUNTIME
220F9B		Number Of Power Pulses
220F9C		ENGRSTPERF ENGRSTRT TMR
221160		CAN_VEH_CFG_C1_STATUS_REGISTER
221161		COMRX_EC_ECUCfg6_Stat
221162		COMRX_EC_Powertrain

In-Plant Test Register

Write ☐ Only display qualified data elements

B[b]	Name	Value	Pending	Units
3[0]	IAC Low RPM Test Status	Passed	Passed	
3[2]	Battery Charging Test Status	Passed	Passed	
3[4]	Vehicle Speed Rationality Test Status	Passed	Passed	
3[6]	Shut Down Timer "Slow" Rationality Test Status	Passed	Passed	
4[0]	Powertrain Control Module Mode	In-Plant	In-Plant	
4[1]	Fuel Prime Test Complete	True	True	
4[2]	Fuel Rail Bleed Test Complete	True	True	
4[3]	Air Pump DTI Actuator Test Complete	True	True	
4[4]	DTI Actuator Test Status	Passed	Passed	
4[6]	IAC High RPM Test Status	Passed	Passed	

Tasto Write



CDA – Read DTC

Read a DTC.

In the section DTCs there's the list of all DTC stored in memory.

With the selection of a DTC it appears all the information related to the DTC: snapshot and extended data record.

The user can compare the environmental data of all DTC

For further details on the DTC it's possible to select the gear wheel.

«Clear DTCs» button can reset the error memory. It's possible to clear the error memory only in KeyOn condition and not in engine running.

CDA 6.10.82
File Edit View Devices Reports Window Help

Device: CANcaseXL (SN:23369)

ECU List PCM Diagnostics

Filter: pcm

Read/Write Data DTCs IO Control On-Board Routines Off-Board Routines ECU Reset PID Editor Read/Write Memory ECU Unlock Calibration

ECU Identification:
Var:7EVer: 14
Selected Database:
PCM-PCM-7E-7E10-000

Single ENV Data Mode Compare ENV Data Mode

Clear DTCs

Code	Description	Status
P0172	Fuel System 1/1 Rich	stored
U05AA	Implausible Data Received From Charge Air Cooler Coolant	stored

Click Here to Filter or Type CTRL+F

Default database matched by variant only

CONFIGURE RE-IDENTIFY ECU

Fuel System 1/1 Rich

Name	Value	Units
Number of DTC	1	
DTC	0172	
Reserved	Undefined	
DTC Readiness Flag	Not Complete	
DTC Storage State	stored	
Warning Indicator Request	ON	
Odometer	1539.0	miles
Accumulation Timer	3	minutes
Ignition Key Cycles	0	
Starts Since Set Counter	0	

Click Here to Filter or Type CTRL+F



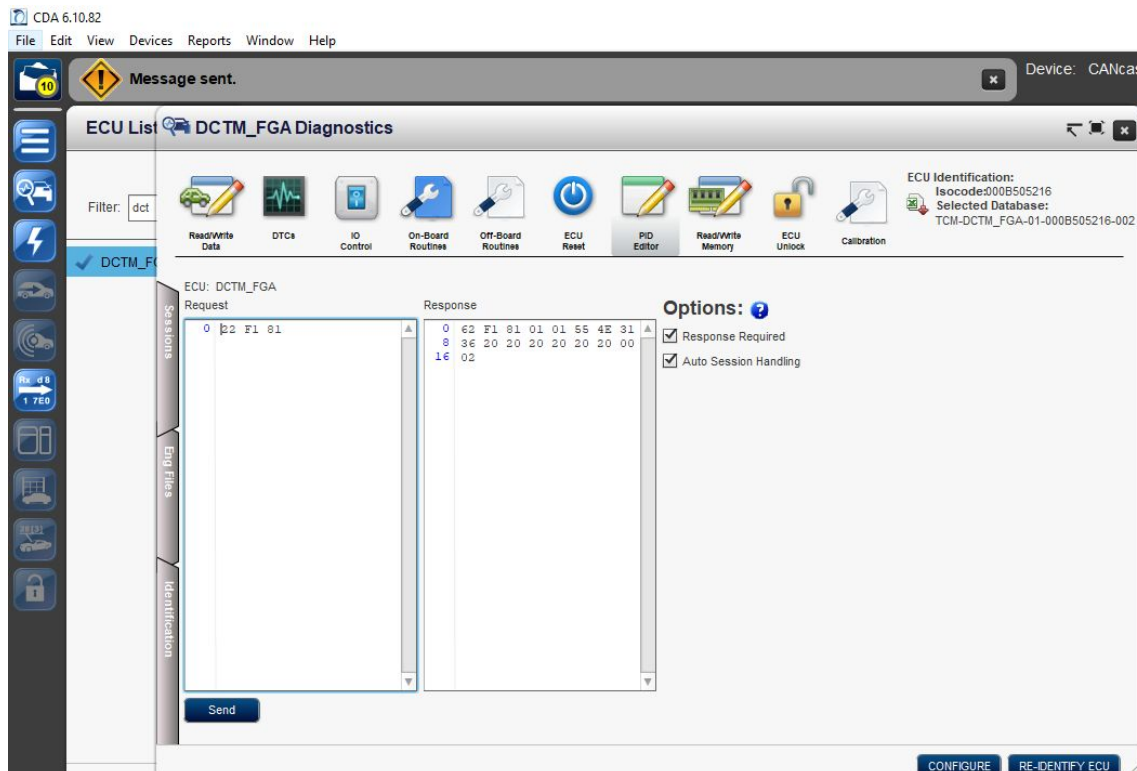
CDA – PID Editor

PID Editor:

In the PID editor section, the user can send a raw request to the ECU and he can read the raw response of the ECU

In the figure, for example the technician required the RDI F1 81.

It's important that the user send the raw data request always from the row 0.





CDA – Flash operation

FLASH operation:

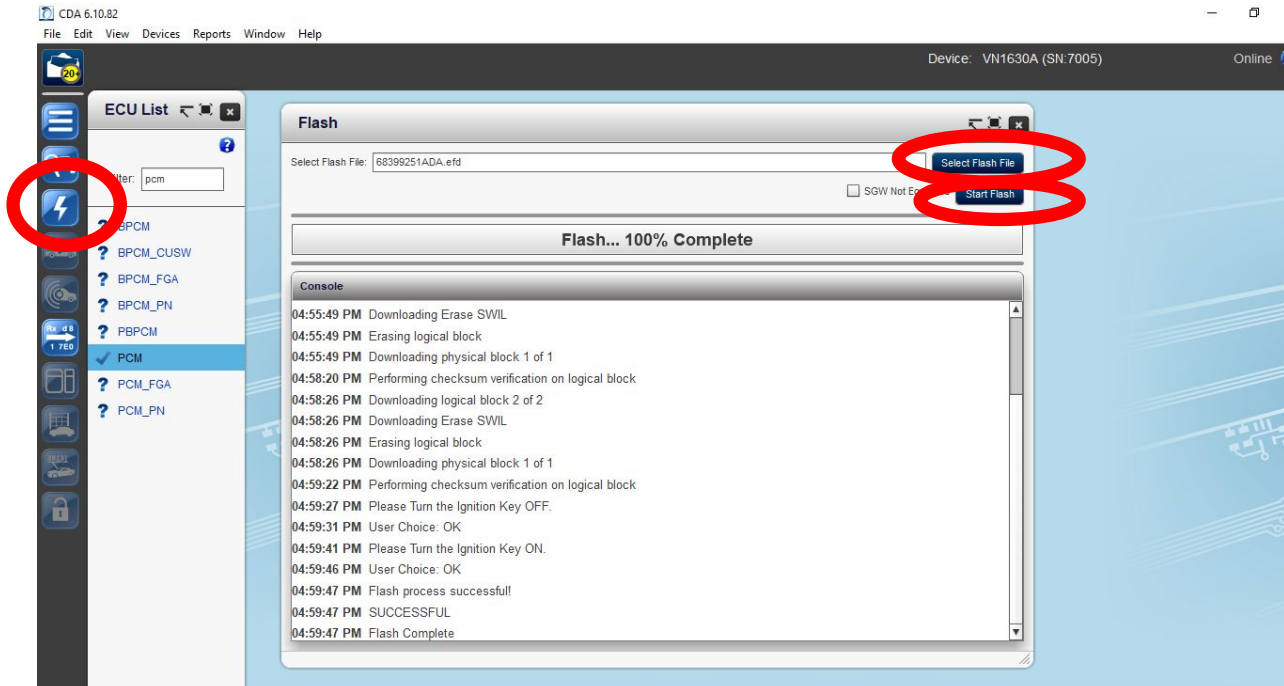
The user can execute a software download on the ECU using the lightning icon on CDA.

It appears the download windows.

The user has to select the file necessary for flashing with a click on «Select Flash File» and with a click on «Start Flash».

It's necessary use a .efd file for the applicative software on the ECU.

CDA will inform the user when the flashing procedure is finished. Usually the download bar locks to 99%; to complete the flash it's necessary a KeyOff->KeyOn procedure.





CDA – Write PROXI

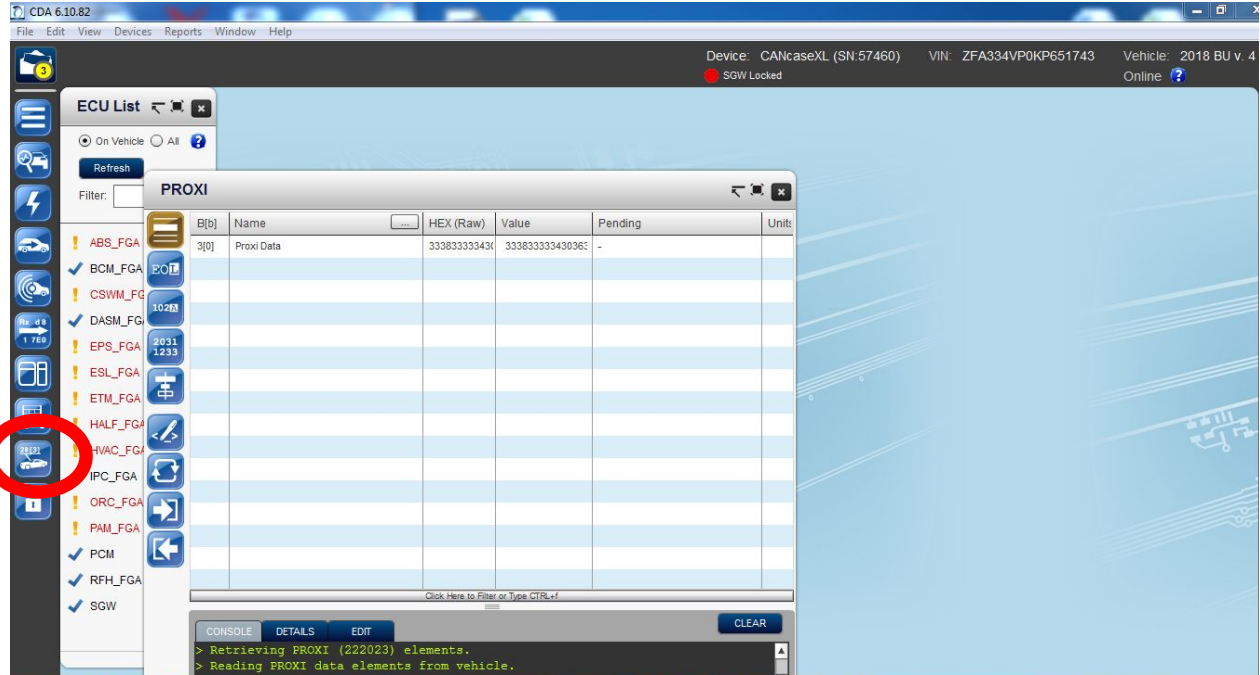
The PROXI procedure with CDA can execute only in Vehicle mode; it's not possible to execute in an HiL bench.

The user has to select the .byt file (configuration file) and he has to execute the PROXI procedure.

The user can modify also the .byt file, but he needs of a specific user privilege.

The PROXI procedure is quite simple in DIAnalyzer instead of CDA.

In both cases, the user need of an user privilege.



PROXI button □ CDA open the first PROXI window and the date are in raw value.



CDA – Write PROXI

The PROXI procedure with CDA can execute only in Vehicle mode; it's not possible to execute in an HiL bench.

The user has to select the .byt file (configuration file) and he has to execute the PROXI procedure.

The user can modify also the .byt file, but he needs of a specific user privilege.

The PROXI procedure is quite simple in DIAAnalyzer instead of CDA.

In both cases, the user need of an user privilege.

CDA 6.10.82

File Edit View Devices Reports Window Help

Device: CANcaseXL (SN:57460) VIN: ZFA334VP0KP651743 Vehicle: 2018 BU v. 4
SGW Locked Online ?

PROXI

B[b]	Name	HEX (Raw)	Value	Units
3[0]	BCM (Body Control Module)	01	Configured as Present by PROXI	
3[1]	ECM (Engine Control Module)	01	Configured as Present by PROXI	
3[2]	EPS (Electric Power Steering)	01	Configured as Present by PROXI	
3[3]	IPC (Instrument Panel Cluster)	01	Configured as Present by PROXI	
3[4]	(reserved)	00	Reserved	
3[5]	RRM (Radio Receiver Module)	00	Configured as Absent by PROXI	
3[6]	BSM/ABS/ASR/EBD/VDC(Brake System Control Module)	01	Configured as Present by PROXI	
3[7]	(reserved)	00	Reserved	
4[0]	(reserved)	00	Reserved	
4[1]	(reserved)	00	Reserved	
4[2]	ACM/MCMMHM (Electronic Climate Control)	01	Configured as Present by PROXI	
4[3]	MTA/TCM/DCTM (Transmission Control Module)	00	Configured as Absent by PROXI	
(reserved)	(reserved)	00	Reserved	
(reserved)	(reserved)	00	Reserved	
4[6]	(reserved)	00	Reserved	
(reserved)	(reserved)	00	Reserved	
4[7]	(reserved)	00	Reserved	
5[0]	(reserved)	00	Reserved	

Click Here to Filter or Type CTRL+F

CONSOLE DETAILS EDIT

Select a value

RAW SET PENDING

Import button, the user can import the PROXI file in .BYT extension.

PROXI data with all configurable parameter (in mode raw and not)



CDA – Write PROXI

The user can execute the PROXI procedure also using the raw data in the PID Editor section.

The screenshot displays the CDA 6.10.82 software interface. The top menu bar includes File, Edit, View, Devices, Reports, Window, and Help. The status bar at the top right shows: Device: CANcaseXL (SN:57460), VIN: ZFA334VP0KP651743, Vehicle: 2018 BU v. 4, Online, and a red indicator for SGW Locked.

The main window is titled "PROXI" and is divided into two main sections. On the left is a table of raw data (hexadecimal values). On the right is a text area for editing the raw transaction.

The raw data table is as follows:

0	2E	20	23	33	38	33	33	34
8	30	36	32	38	38	32	4F	55
16	54	50	55	54	2D	53	49	54
24	20	18	01	22	4F	04	20	45
32	10	00	01	02	02	00	00	00
40	00	00	01	00	4F	04	20	45
48	10	00	01	02	02	00	00	00
56	00	00	00	00	D1	FA	34	81
64	AD	81	08	02	B0	27	09	07
72	40	10	10	01	A3	1E	04	92
80	01	35	73	14	01	4F	03	B4
88	18	A3	A8	01	5D	02	05	00
96	92	00	10	25	00	00	30	01
104	00	25	12	54	EB	64	06	A5
112	00	52	00	01	8A	10	3D	40
120	10	03	03	C5	00	D5	24	13
128	0A	00	49	00	81	14	7B	42
136	05	00	00	13	A2	99	71	20
144	80	A4	82	01	00	00	05	00
152	00	00	00	00	00	00	00	01
160	0B	C0	01	03	82	00	01	00
168	00	00	02	00	00	00	00	00
176	00	00	00	00	00	00	00	00
184	00	00	00	00	00	00	00	00
192	00	00	00	00	00	00	00	00
200	00	00	00	00	00	00	00	00
208	00	00	00	00	00	00	00	00
216	00	00	00	00	00	00	00	00
224	00	00	00	00	00	00	00	00
232	00	00	00	00	00	00	00	00

The right section is titled "EDIT the raw transaction that will be sent." and contains the text: "Changes to the Data Display will not replace the contents of the PID Editor." Below this is a large text area for editing. At the bottom right, there are "SAVE" and "LOAD" buttons.

A red circle highlights the "2031 1233" button in the left sidebar, which is part of the "PID Editor" section.

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